

# Chapter 5: Comparison of Several Mean vectors (Part 2)

Zhixiang ZHANG

University of Macau

zhixzhang@um.edu.mo

March 5, 2025

# Outline

Following our approach to the one-way MANOVA, we shall start from a univariate two-way fixed-effects model and then simply generalize to the multivariate case by analogy.

# Two-way ANOVA

|                |          | Factor B              |                       | no. groups |                       |
|----------------|----------|-----------------------|-----------------------|------------|-----------------------|
|                |          | 1                     | 2                     | ...        | $b$                   |
| Factor A       | 1        | $\{X_{11r}\}_{r=1}^n$ | $\{X_{12r}\}_{r=1}^n$ | ...        | $\{X_{1br}\}_{r=1}^n$ |
|                | 2        | $\{X_{21r}\}_{r=1}^n$ | $\{X_{22r}\}_{r=1}^n$ | ...        | $\{X_{2br}\}_{r=1}^n$ |
|                | $\vdots$ | $\vdots$              | $\vdots$              |            | $\vdots$              |
| no. groups $g$ |          | $\{X_{g1r}\}_{r=1}^n$ | $\{X_{g2r}\}_{r=1}^n$ | ...        | $\{X_{gbr}\}_{r=1}^n$ |

- Observation:  $X_{lkr}$  = the  $r^{th}$  <sup>for observation</sup> observation at level  $l$  of Factor A and level  $k$  of Factor B.
- Factor A:  $l = 1, \dots, g$
- Factor B:  $k = 1, \dots, b$
- Replications:  $r = 1, \dots, n$  For now, a **balanced** design, which means the sample sizes are equal across all factor(treatment) combinations.

$$I_A = I_B = r$$

# ANOVA model for two factors

The mean of the response is modeled as

$$\underbrace{\mu_{lk}} = \underbrace{\mu}_{\text{overall level}} + \underbrace{\tau_l}_{\substack{\text{fixed effect} \\ \text{of factor A} \\ \text{at level } l}} + \underbrace{\beta_k}_{\substack{\text{fixed effect} \\ \text{of factor B} \\ \text{at level } k}} + \underbrace{\gamma_{lk}}_{\substack{\text{interaction between} \\ \text{factor A and B} \\ \text{at level } l \& k}}$$

if  $\gamma = \mathbb{I}k = 0$ , it is additive  
if  $\gamma = \mathbb{I}k \neq 0$ , it is not additive

where

$$\sum_{l=1}^g \tau_l = \sum_{k=1}^b \beta_k = \sum_{l=1}^g \gamma_{lk} = \sum_{k=1}^b \gamma_{lk} = 0$$

The effects are not additive; there is an interaction.

Correspondingly,  $X_{lkr} = \mu + \tau_l + \beta_k + \gamma_{lk} + \epsilon_{lkr}$ .

# An illustration of interactive effect

addition across A across B

$$\mu_{AB} = \mu + \rho_A + \mu_B$$

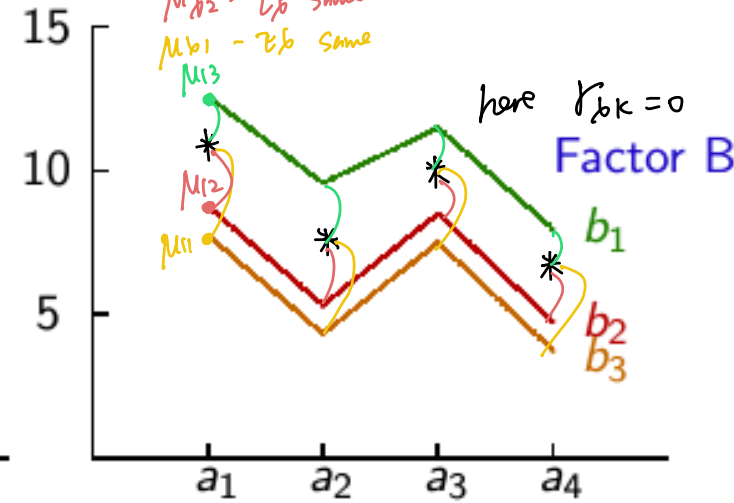
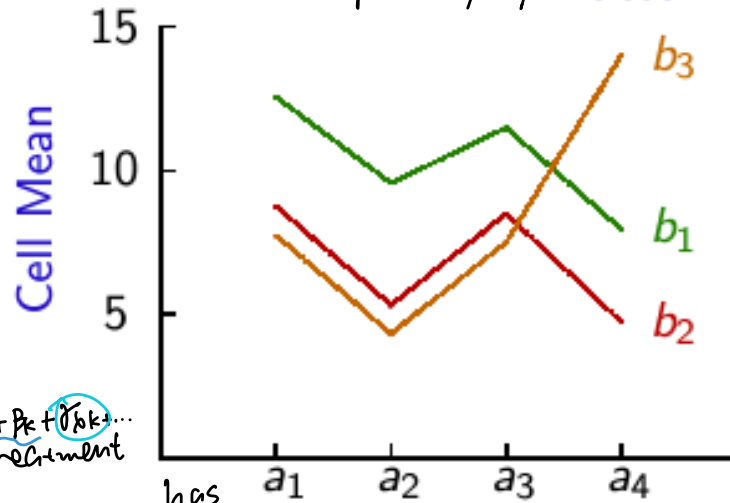
$$\mu_{AB} - \mu_A = \mu + \rho_B$$

$$\mu_{b|k} = \mu + \tau_b + \beta_k + \gamma_{bk}$$

$$\mu_{b|k} - \tau_b = \underbrace{\mu + \beta_k}_{\text{same}} + \gamma_{bk}$$

0:  $\mu_{b|k} - \tau_b$  doesn't depend on  $b$  index;  
not 0:  $\mu_{b|k} - \tau_b$  depends on  $b$  index.

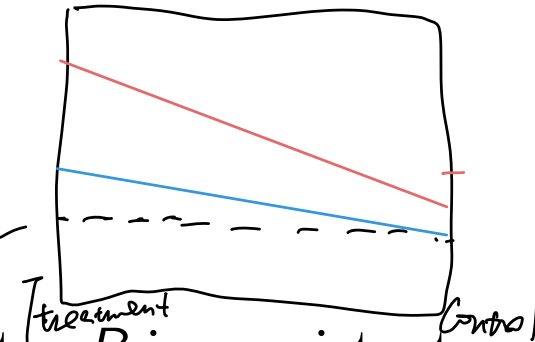
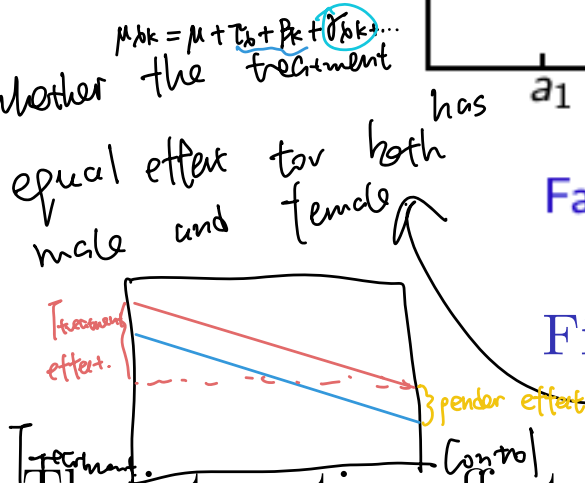
have  $\gamma_{bk} \neq 0$



Factor A

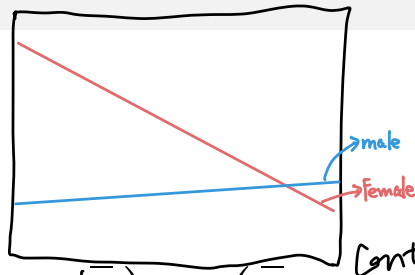
Factor A

Figure: Interaction & No interaction



The interaction effect determines whether the effect of factor  $B$  is consistent across the different levels of factor  $A$  (vice versa). In other words, no interaction means that for any fixed level  $k$  of factor  $B$ , the difference  $\mu_{b|k} - \tau_b = \beta_k$  should remain constant across different levels  $\ell$  of factor  $A$ .

# Observation as Sum of Means



$$X_{lkr} = \underbrace{\bar{X}}_{\hat{\mu}} + \underbrace{(\bar{X}_{l..} - \bar{X})}_{\hat{\tau}_l} + \underbrace{(\bar{X}_{.k.} - \bar{X})}_{\hat{\beta}_k} + \underbrace{(\bar{X}_{lk.} - \bar{X}_{l..} - \bar{X}_{.k.} + \bar{X})}_{\hat{\gamma}_{lk}} + \underbrace{(X_{lkr} - \bar{X}_{lk.})}_{\hat{\epsilon}_{lkr}}$$

$$\mu_{lk} = \mu + \tau_l + \beta_k + \gamma_{lk}$$

$$\mu_{lk} - \tau_l = \mu + \beta_k + \gamma_{lk} \quad (= \mu + \beta_k)$$

If  $\gamma_{lk} = 0$ , then  $\mu_{lk} - \tau_l$  are the same for different  $l$

For any fixed level  $k$  of factor B, the difference  $\mu_{lk} - \tau_l$  should remain constant across different level  $l$  of factor A

where

- $\bar{X}$  = overall sample mean.  $= \frac{1}{n \cdot g \cdot b} \sum_{k=1}^b \sum_{l=1}^g \sum_{r=1}^n X_{lkr}$
- $\bar{X}_{l..}$  = mean for  $l^{\text{th}}$  level of Factor A (or “row” mean).  $= \frac{1}{nb} \sum_{k=1}^b \sum_{r=1}^n X_{lkr}$
- $\bar{X}_{.k.}$  = mean for  $k^{\text{th}}$  level of Factor B (or “column” mean).  $= \frac{1}{ng} \sum_{l=1}^g \sum_{r=1}^n X_{lkr}$
- $\bar{X}_{lk.}$  = mean for the  $l^{\text{th}}$  level of Factor A and  $k^{\text{th}}$  level of Factor B (or the “cell” mean).  $= \frac{1}{n} \sum_{r=1}^n X_{lkr}$
- The decomposition of the observations above shows the estimates of the population effects.

After decomposing sum of squares, we have some terms like

$$\sum_{l=1}^g \sum_{k=1}^b \sum_{i=1}^n (\bar{x}_{li} - \bar{x})(\bar{x}_{ki} - \bar{x})$$

$$= n \sum_{k=1}^b \left( \sum_{l=1}^g (\bar{x}_{lk} - \bar{x}) \right) (\bar{x}_{k.} - \bar{x}) = n \sum_{k=1}^b 0 \cdot (\bar{x}_{k.} - \bar{x}) = 0$$

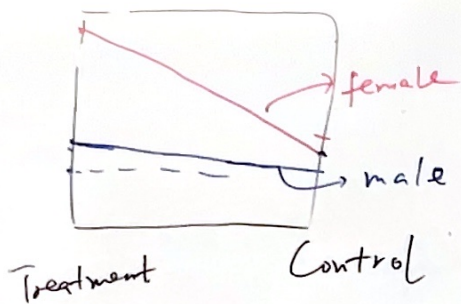
Factor 1

$$n \sum_{l=1}^g \sum_{k=1}^b (\bar{x}_{li} - \bar{x})(\bar{x}_{lik} - \bar{x}_{li} - \bar{x}_{k.} + \bar{x}) = 0$$

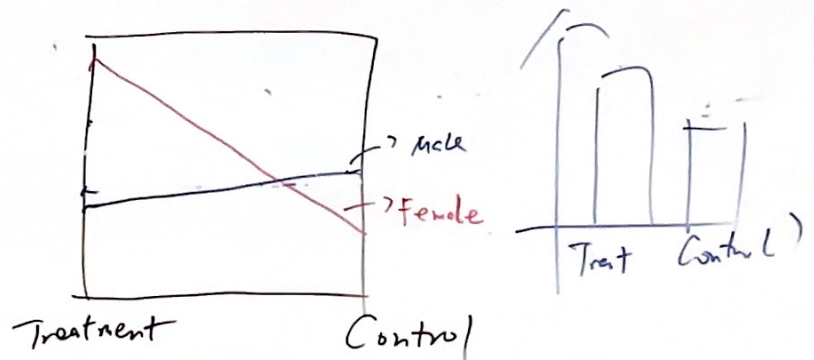
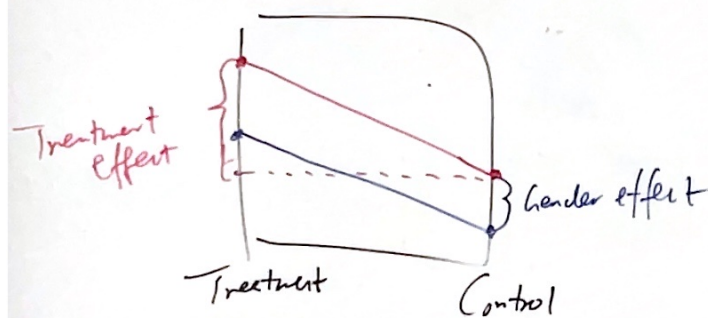
$$= \sum_{l=1}^g \sum_{k=1}^b (\bar{x}_{lk} - \bar{x})(\bar{x}_{lik} - \bar{x}_{li} - \bar{x}_{k.} + \bar{x})$$

$$= \sum_{l=1}^g (\bar{x}_{l..} - \bar{x}) \left( \sum_{k=1}^b (\bar{x}_{lik} - \bar{x}_{li} - \bar{x}_{k.} + \bar{x}) \right) + 0$$

$$= \sum_{l=1}^g (\bar{x}_{l..} - \bar{x}) \cdot 0 + 0 = 0$$



whether the treatment has equal effect for both male and female?



we cannot say treatment is effect or not in general  
depend on gender

The

# Decomposition of Sums of Squares

$$X_{lkr} = \mu_{lkr} + \epsilon_{lkr}$$

$$\mu_{lkr} = \underbrace{\bar{X}}_{\hat{\mu}} + \underbrace{(\bar{X}_{l..} - \bar{X})}_{\hat{\tau}_l} + \underbrace{(\bar{X}_{.k.} - \bar{X})}_{\hat{\beta}_k} + \underbrace{(\bar{X}_{l.k.} - \bar{X}_{l..} - \bar{X}_{.k.} + \bar{X})}_{\hat{\gamma}_{lk}} + \underbrace{(X_{lkr} - \bar{X}_{l.k.})}_{\hat{\epsilon}_{lkr}}$$

To get the sums of squares: *all [ ] means Info*

*[ $\mu + A + B + \text{Interact}(A, B)$ ] - [ $\mu + A$ ] - [ $\mu + B$ ] + [ $\mu$ ]*

- Subtract  $\bar{X}$  from both sides; then square both sides.
- Sum over observations within cells, levels of Factor A, and levels of Factor B.

After a little algebra, The sum of squares decomposition is

$$\sum_{l=1}^g \sum_{k=1}^b \sum_{r=1}^n (X_{lkr} - \bar{X})^2 = \sum_{l=1}^g bn (\bar{X}_{l..} - \bar{X})^2 + \sum_{k=1}^b gn (\bar{X}_{.k.} - \bar{X})^2$$

$$+ \sum_{l=1}^g \sum_{k=1}^b n (\bar{X}_{l.k.} - \bar{X}_{l..} - \bar{X}_{.k.} + \bar{X})^2 + \sum_{l=1}^g \sum_{k=1}^b \sum_{r=1}^n (X_{lkr} - \bar{X}_{l.k.})^2$$



# Sums of Squares decomposition

We get an orthogonal decomposition of the sums of squares:

$$SS_{\text{total}} = SS_A + SS_B + SS_{AB} + SS_{\text{residual}}$$

Note:  $SS_{\text{total}}$  is corrected for the grand mean. Also, we get a decomposition of degrees of freedom,

$$\underbrace{(gbn - 1)}_{\text{total}} = \underbrace{(g - 1)}_{\text{Factor A}} + \underbrace{(b - 1)}_{\text{Factor B}} + \underbrace{(g - 1)(b - 1)}_{\text{Interaction}} + \underbrace{gb(n - 1)}_{\text{Residual}}$$

# ANOVA summary table

Two-way ANOVA Table

| Source of variation | Degrees of freedom | SS<br>Sums of Squared                                                                               |
|---------------------|--------------------|-----------------------------------------------------------------------------------------------------|
| Factor A            | $(g - 1)$          | $SS_A = \sum_{l=1}^g bn (\bar{X}_{l..} - \bar{X})^2$                                                |
| Factor B            | $(b - 1)$          | $SS_B = \sum_{k=1}^b gn (\bar{X}_{.k.} - \bar{X})^2$                                                |
| Interaction         | $(g - 1)(b - 1)$   | $SS_{AB} = \sum_{l=1}^g \sum_{k=1}^b n (\bar{X}_{lk.} - \bar{X}_{l..} - \bar{X}_{.k.} + \bar{X})^2$ |
| Residual<br>(error) | $gb(n - 1)$        | $SS_{\text{res}} = \sum_{l=1}^g \sum_{k=1}^b \sum_{r=1}^n (X_{lkr} - \bar{X}_{lk.})^2$              |
| Total               | $gbn - 1$          | $SS_{\text{total}} = \sum_{l=1}^g \sum_{k=1}^b \sum_{r=1}^n (X_{lkr} - \bar{X})^2$                  |

# Hypothesis tests

- Interaction:  $H_0 : \gamma_{11} = \gamma_{12} = \cdots = \gamma_{gb} = 0$ . If the null is true and all assumptions valid,

$$F = \frac{SS_{AB}/((g-1)(b-1))}{SS_{\text{res}}/(gb(n-1))} \sim \mathcal{F}_{(g-1)(b-1), gb(n-1)}$$

- Factor A:  $H_0 : \tau_1 = \tau_2 = \cdots = \tau_g$ . If the null is true and all assumptions valid,

$$F = \frac{SS_A/(g-1)}{SS_{\text{res}}/(gb(n-1))} \sim \mathcal{F}_{(g-1), gb(n-1)}$$

- Factor B:  $H_0 : \beta_1 = \beta_2 = \cdots = \beta_b$ . If the null is true and all assumptions valid,

$$= \frac{SS_B/(b-1)}{SS_{\text{res}}/(gb(n-1))} \sim \mathcal{F}_{(b-1), gb(n-1)}$$

# Two-way MANOVA

For  $p$  variables,  $\mathbf{x}_{lkr}$  is a  $p \times 1$  vector of measurements on  $p$  variables. The model

$$\mathbf{x}_{lkr} = \boldsymbol{\mu} + \boldsymbol{\tau}_l + \boldsymbol{\beta}_k + \boldsymbol{\gamma}_{lk} + \boldsymbol{\epsilon}_{lkr}$$

and  $\boldsymbol{\epsilon}_{lkr} \sim \mathcal{N}_p(\mathbf{0}, \boldsymbol{\Sigma})$  and independent.

- $l = 1, \dots, g$
- $k = 1, \dots, b$
- $r = 1, \dots, n_{lk}$
- and for identification

$$\sum_{l=1}^g \boldsymbol{\tau}_l = \sum_{k=1}^b \boldsymbol{\beta}_k = \sum_{l=1}^g \boldsymbol{\gamma}_{lk} = \sum_{k=1}^b \boldsymbol{\gamma}_{lk} = \mathbf{0}$$

# Decomposition of SS

$$\begin{aligned}
 & \underbrace{\sum_l \sum_k \sum_r (\bar{\mathbf{x}}_{l..} - \bar{\mathbf{x}}) (\bar{\mathbf{x}}_{l..} - \bar{\mathbf{x}})^\top}_{SS_{\text{total}}} \\
 &= \underbrace{\sum_l bn (\bar{\mathbf{x}}_{l..} - \bar{\mathbf{x}}) (\bar{\mathbf{x}}_{l..} - \bar{\mathbf{x}})^\top}_{SS_A} + \underbrace{\sum_k gn (\bar{\mathbf{x}}_{.k.} - \bar{\mathbf{x}}) (\bar{\mathbf{x}}_{.k.} - \bar{\mathbf{x}})^\top}_{SS_B} \\
 &+ \underbrace{\sum_l \sum_k n (\bar{\mathbf{x}}_{l/k.} - \bar{\mathbf{x}}_{l..} - \bar{\mathbf{x}}_{.k.} + \bar{\mathbf{x}}) (\bar{\mathbf{x}}_{l/k.} - \bar{\mathbf{x}}_{l..} - \bar{\mathbf{x}}_{.k.} + \bar{\mathbf{x}})^\top}_{SS_{\text{interaction}}} \\
 &+ \underbrace{\sum_l \sum_k \sum_r (\mathbf{x}_{lkr} - \bar{\mathbf{x}}_{l/k.}) (\mathbf{x}_{lkr} - \bar{\mathbf{x}}_{l/k.})^\top}_{SS_{\text{residual}}}
 \end{aligned}$$

# Tow-way MANOVA summary table

Two-way MANOVA Table

| Source of variation | Degrees of freedom | SS<br>Sums of Squared                                                                                                                    |
|---------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Factor A            | $(g - 1)$          | $SS_A = \sum_{l=1}^g bn (\bar{\mathbf{x}}_{l..} - \bar{\mathbf{x}})^2$                                                                   |
| Factor B            | $(b - 1)$          | $SS_B = \sum_{k=1}^b gn (\bar{\mathbf{x}}_{.k.} - \bar{\mathbf{x}})^2$                                                                   |
| Interaction         | $(g - 1)(b - 1)$   | $SS_{AB} = \sum_{l=1}^g \sum_{k=1}^b n (\bar{\mathbf{x}}_{l k.} - \bar{\mathbf{x}}_{l..} - \bar{\mathbf{x}}_{.k.} + \bar{\mathbf{x}})^2$ |
| Residual<br>(error) | $gb(n - 1)$        | $SS_{\text{res}} = \sum_{l=1}^g \sum_{k=1}^b \sum_{r=1}^n (\mathbf{x}_{lkr} - \bar{\mathbf{x}}_{l k.})^2$                                |
| Total               | $gbn - 1$          | $SS_{\text{total}} = \sum_{l=1}^g \sum_{k=1}^b \sum_{r=1}^n (\mathbf{x}_{lkr} - \bar{\mathbf{x}})^2$                                     |

# Hypothesis Testing: Interaction First

most important  
if  $\gamma \neq 0$ , then testing  
main effect is meaningless

Test  $H_0 : \gamma_{11} = \gamma_{12} = \dots = \gamma_{gb} = 0$  first  
then  $H_0 : \tau_1 = \dots = \tau_g = 0$  and  $H_0 : \beta_1 = \dots = \beta_b = 0$

- Test Statistic is Wilk's Lambda

$$\Lambda^* = \frac{|SS_{\text{residual}}|}{|SS_{\text{residual}} + SS_{\text{interaction}}|}$$

$\Lambda^*$  is small when  $SS_{\text{interaction}}$  is large.

- Let  $\nu_h = \nu_{\text{interaction}} = (g-1)(b-1)$ , and  $\nu_e = \nu_{\text{residual}} = gb(n-1)$
- Distribution of  $\Lambda^*$  are given on the following slide for various cases.
- If you don't have  $p \leq 2$  or  $\nu_h \leq 2$ , but you have large  $n$  then the approximate distribution of

$$\left( \nu_e - \frac{p+1-\nu_h}{2} \right) \ln(\Lambda^*) \quad \text{is} \quad \chi^2_{p\nu_h}$$

# Distribution of Wilk's Lambda

$$\text{Wilk's } \Lambda^* = \frac{|SS_{\text{residual}}|}{|SS_{\text{residual}} + SS_{\text{interaction}}|}$$

| Number<br>variables | <i>df</i> for<br>Hypothesis | Sampling distribution                                                                                                                       |
|---------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| $p = 1$             | $\nu_h \geq 1$              | $\left(\frac{\nu_e}{\nu_h}\right) \left(\frac{1-\Lambda^*}{\Lambda^*}\right) \sim \mathcal{F}_{\nu_h, \nu_e}$                               |
| $p = 2$             | $\nu_h \geq 1$              | $\left(\frac{\nu_e-1}{\nu_h}\right) \left(\frac{1-\sqrt{\Lambda^*}}{\sqrt{\Lambda^*}}\right) \sim \mathcal{F}_{2\nu_h, 2(\nu_e-1)}$         |
| $p \geq 1$          | $\nu_h = 1$                 | $\left(\frac{\nu_e+\nu_h-p}{p}\right) \left(\frac{1-\Lambda^*}{\Lambda^*}\right) \sim \mathcal{F}_{p, (\nu_e+\nu_h-p)}$                     |
| $p \geq 2$          | $\nu_h = 2$                 | $\left(\frac{\nu_e+\nu_h-p-1}{p}\right) \left(\frac{1-\sqrt{\Lambda^*}}{\sqrt{\Lambda^*}}\right) \sim \mathcal{F}_{2p, 2(\nu_e+\nu_h-p-1)}$ |

where  $\nu_h$  = degrees of freedom for hypothesis, and  $\nu_e$  = degrees of freedom for error (residual).



# Hypothesis Testing: Interaction first

You should test the interaction first.

- Interpretation: The interaction effect determines whether the effect of factor  $A$  (eg. gender) is consistent across the different levels of factor  $B$  (eg. different interventions).
- If  $H_0$  for the interaction is rejected, then you don't do tests for main effects because the Factor main effects do not have a clear interpretation.
- J&W recommend doing p-univariate 2-way ANOVAs to find out “where” the interactions exist. (i.e., are there interactions for just some of the  $p$  variables and not others?). For variables with no interaction, you can interpret main effects.
- In general, if you retain the null hypothesis for interaction, we continue and test main effects...

# Hypothesis testing: Main effects

Factor A:  $H_0 : \tau_1 = \tau_2 = \cdots = \tau_g = \mathbf{0}$  versus  $H_1 : \text{at least one } \tau_l \neq \mathbf{0}$

$$\Lambda_A^* = \frac{|SS_{\text{residual}}|}{|SS_{\text{residual}} + SS_A|}$$

and  $\nu_h = g - 1$ . For  $p \leq 2$  or  $\nu_A \leq 2$  (i.e.,  $g \leq 3$ ), use the exact distribution given in the table.

Factor B:  $H_0 : \beta_1 = \beta_2 = \cdots = \beta_b = \mathbf{0}$  versus  $H_1 : \text{at least one } \beta_k \neq \mathbf{0}$

$$\Lambda_B^* = \frac{|SS_{\text{residual}}|}{|SS_{\text{residual}} + SS_B|}$$

and  $\nu_h = b - 1$ . For  $p \leq 2$  or  $\nu_B \leq 2$  (i.e.,  $b \leq 3$ ), use the exact distribution given in the table.

For large  $n$ :

$$- \left( \nu_e - \frac{p + 1 - \nu_h}{2} \right) \ln(\Lambda_h^*) \approx \chi_{p\nu_h}^2$$

# Unbalanced design

The order of decomposition of SS will have an impact on the result, so this is more tricky, not it will not be considered in this course.

# Example

The optimum conditions for extruding plastic film have been examined using a technique called Evolutionary Operation.

Three responses tear resistance, gloss, and opacity-were measured at two levels of the factors, rate of extrusion and amount of an additive. The measurements were repeated  $n = 5$  times at each combination of the factor levels.

| $x_1$ = tear resistance, $x_2$ = gloss, and $x_3$ = opacity |            |                              |                   |                   |                   |                   |                   |
|-------------------------------------------------------------|------------|------------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|                                                             |            | Factor 2: Amount of additive |                   |                   |                   |                   |                   |
|                                                             |            | Low (1.0%)                   |                   |                   | High (1.5%)       |                   |                   |
| Factor 1: Change<br>in rate of extrusion                    | Low (-10)% | $\underline{x_1}$            | $\underline{x_2}$ | $\underline{x_3}$ | $\underline{x_1}$ | $\underline{x_2}$ | $\underline{x_3}$ |
|                                                             |            | [6.5                         | 9.5               | 4.4]              | [6.9              | 9.1               | 5.7]              |
|                                                             |            | [6.2                         | 9.9               | 6.4]              | [7.2              | 10.0              | 2.0]              |
|                                                             |            | [5.8                         | 9.6               | 3.0]              | [6.9              | 9.9               | 3.9]              |
|                                                             |            | [6.5                         | 9.6               | 4.1]              | [6.1              | 9.5               | 1.9]              |
|                                                             | High (10%) | [6.5                         | 9.2               | 0.8]              | [6.3              | 9.4               | 5.7]              |
|                                                             |            | $\underline{x_1}$            | $\underline{x_2}$ | $\underline{x_3}$ | $\underline{x_1}$ | $\underline{x_2}$ | $\underline{x_3}$ |
|                                                             |            | [6.7                         | 9.1               | 2.8]              | [7.1              | 9.2               | 8.4]              |
|                                                             |            | [6.6                         | 9.3               | 4.1]              | [7.0              | 8.8               | 5.2]              |
|                                                             |            | [7.2                         | 8.3               | 3.8]              | [7.2              | 9.7               | 6.9]              |
|                                                             |            | [7.1                         | 8.4               | 1.6]              | [7.5              | 10.1              | 2.7]              |
| [6.8                                                        | 8.5        | 3.4]                         | [7.6              | 9.2               | 1.9]              |                   |                   |

Figure: Plastic film data

# Data in R

```
> Plastic
```

|    | tear | gloss | opacity | rate | additive |
|----|------|-------|---------|------|----------|
| 1  | 6.5  | 9.5   | 4.4     | Low  | Low      |
| 2  | 6.2  | 9.9   | 6.4     | Low  | Low      |
| 3  | 5.8  | 9.6   | 3.0     | Low  | Low      |
| 4  | 6.5  | 9.6   | 4.1     | Low  | Low      |
| 5  | 6.5  | 9.2   | 0.8     | Low  | Low      |
| 6  | 6.9  | 9.1   | 5.7     | Low  | High     |
| 7  | 7.2  | 10.0  | 2.0     | Low  | High     |
| 8  | 6.9  | 9.9   | 3.9     | Low  | High     |
| 9  | 6.1  | 9.5   | 1.9     | Low  | High     |
| 10 | 6.3  | 9.4   | 5.7     | Low  | High     |
| 11 | 6.7  | 9.1   | 2.8     | High | Low      |
| 12 | 6.6  | 9.3   | 4.1     | High | Low      |
| 13 | 7.2  | 8.3   | 3.8     | High | Low      |
| 14 | 7.1  | 8.4   | 1.6     | High | Low      |
| 15 | 6.8  | 8.5   | 3.4     | High | Low      |
| 16 | 7.1  | 9.2   | 8.4     | High | High     |
| 17 | 7.0  | 8.8   | 5.2     | High | High     |
| 18 | 7.2  | 9.7   | 6.9     | High | High     |
| 19 | 7.5  | 10.1  | 2.7     | High | High     |
| 20 | 7.6  | 9.2   | 1.9     | High | High     |

# Test interaction effects

To test for interaction, we compute

$$\Lambda^* = \frac{|\text{SS}_{\text{residual}}|}{|\text{SS}_{\text{residual}} + \text{SS}_{\text{interaction}}|} = \frac{275.7098}{354.7906} = .7771.$$

Since  $p = 3$ ,  $\nu_h = 1$ , and  $\nu_e = 16$ , use

$$\frac{\nu_e + \nu_h - p}{p} \frac{1 - \Lambda^*}{\Lambda^*} = \frac{14}{3} \left( \frac{1 - .7771}{.7771} \right) = 1.34,$$

and it is smaller than  $F_{3,14}(.05) = 3.34$ . Therefore we do not reject the hypothesis  $H_0 : \gamma_{11} = \gamma_{12} = \gamma_{21} = \gamma_{22} = \mathbf{0}$  (no interaction effects).

## Test main effects

To test for the effect of factor A (rate of extrusion), i.e.,  $H_0 : \boldsymbol{\tau}_1 = \boldsymbol{\tau}_2 = \mathbf{0}$ , we calculate

$$\Lambda_A^* = \frac{|\text{SS}_{\text{residual}}|}{|\text{SS}_{\text{residual}} + \text{SS}_A|} = \frac{275.7008}{722.0212} = .3819$$

Since  $\nu_h$  for this test is one (factor A has two levels,  $\nu_h = g - 1 = 1$ ), use

$$F_A = \frac{\nu_e + \nu_h - p}{p} \left( \frac{1 - \Lambda_A^*}{\Lambda_A^*} \right) \sim \mathcal{F}_{p, (\nu_e + \nu_h - p)}$$

and we find  $F_A = 7.55 > F_{3,14}(.05) = 3.34$ . So we reject  $H_0$  and conclude that change in rate of extrusion affects the responses.

# Conclusion

Similarly, we can test the effect of amount of additive and conclude that it also affects the responses.

The final conclusion is, both factors affect the responses, and they do so in an additive manner.